

Self-Growth Program

DTTP Full-Stack Developer Self-Growth Roadmap

Ten learning levels that turn full-stack reference material into a practical Deshmukh Technologies growth path — from web fundamentals to professional engineering.

PURPOSE

Self-directed full-stack growth

TRACKS

Java / Spring Boot or Python / FastAPI

PRACTICE STACK

SQL + GitHub + Docker + CI/CD + AWS

LEVELS

10 progressive stages

FRONTEND

React + TypeScript

COMPANION

90-Day DTTP Handbook

Learn. Build. Solve. Review. Deploy.
Grow.

INTERNAL TALENT DEVELOPMENT
STANDARD

DOCUMENT DTTP Self-Growth Program	VERSION 1.0	CLASSIFICATION Internal Training Standard	OWNER Deshmukh Technologies
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ROADMAP

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00 How to use this program

Full-stack reference material provides a strong foundation. It is **not** the entire DTTP curriculum. Use it as the **Full-Stack Knowledge Reference**, then apply it through Deshmukh Technologies tracks and real projects.

Do not ask trainees to study the reference as one large syllabus. Organize growth into ten levels, then combine each level with Java/Spring Boot or Python/FastAPI, React/TypeScript, SQL, GitHub, Docker, CI/CD, AWS, and real Deshmukh Technologies work.

01 Web Fundamentals

02 Frontend Fundamentals

03 React + TypeScript

04 HTTP + REST + Web APIs

05 Java / Python Backend

06 Databases

07 Architecture

08 Testing + Security + Performance

09 Git + Docker + CI/CD + Cloud

10 Agile + Professional Engineering

The destination is a **self-directed full-stack developer** who can build, test, debug, explain, document, and maintain software.

01 Level 1 — Web Fundamentals

These are the starting concepts for understanding full-stack development.

Client & server

URLs, domains, IP addresses

Web application structure

HTML

CSS

JavaScript

Server-side logic

Full-stack concepts

Development tools

A trainee who cannot explain how a browser request reaches a server is not ready for React or Spring Boot.

02 Level 2 — Frontend Engineering

Progress from HTML, CSS and JavaScript into the DOM, Fetch/API usage, and accessibility.

STRUCTURE & STYLE

- HTML elements and forms
- CSS selectors and specificity
- Flexbox
- CSS Grid

BEHAVIOR & ACCESS

- JavaScript variables, types, functions
- Error handling, objects, arrays
- DOM, Fetch API, Web APIs
- Accessibility

03 Level 3 — React & Modern Frontend

The common DTTP frontend path is **React + TypeScript**.

Components

State

Lifecycle

Forms

Component hierarchy

Context API

Routing

Responsive design

SPA architecture

04 Level 4 — Web Communication & APIs

These topics teach what actually happens between frontend and backend.

HTTP

- Requests / responses
- Headers
- Methods
- Status codes
- Cookies

Realtime

- Polling
- Server-Sent Events
- WebSockets

API styles

- JSON
- REST
- GraphQL

05 Level 5 — Backend Development

Specialize server-side concepts into the Deshmukh Technologies tracks. Trainees are not required to learn every backend language in the reference material.

JAVA TRACK

JAVA

SPRING BOOT

REST

JPA / HIBERNATE

SPRING SECURITY

PYTHON TRACK

PYTHON

FASTAPI

PYDANTIC

SQLALCHEMY

AUTHENTICATION

06 Level 6 — Database Engineering

This is a major self-growth area. **SQL + MySQL / PostgreSQL is mandatory**. NoSQL is introduced later.

RELATIONAL — MANDATORY

- Tables and relationships
- SQL
- ORM
- Transactions
- Query design

NON-RELATIONAL — LATER

- Key-value
- Document
- Graph
- Column-oriented databases

ORM knowledge must not replace SQL knowledge.

07 Level 7 — Architecture

This is where trainees move from **developer thinking to engineer thinking**. Study architectural models, but understand **why** architecture changes.

Layered architecture

Client-server

Multi-tier

Monolith

SOA

Microservices

Component-based

Microfrontends

Messaging

MVC

MVP

MVVM

DTTP ARCHITECTURE PRINCIPLE

Do not teach microservices first. Teach Monolith → Layered Architecture → Modular Monolith → Distributed Systems → Microservices.

08 Level 8 — Testing, Security & Performance

These three become permanent self-growth subjects.

Testing

- Automated testing
- Unit and integration tests
- TDD and coverage
- Test doubles, spies, stubs, mocks

Security

- Injection, XSS, access control
- Authentication and authorization
- Sensitive data, TLS / HTTPS
- CORS, CSP, token authentication

Performance

- Measure before optimizing
- Connection and image optimization
- Server- and client-side caching
- Minification, compression, lazy loading, preloading

Measure performance before optimizing it.

09 Level 9 — DevOps & Deployment

This level connects directly to the DTTP development workflow.

Git

- Repository and staging
- Commits
- Remote repositories
- Branches and merging

Docker

- Images
- Containers
- Dockerfiles
- Docker Compose

Deployment

- Build and deploy
- Hosting and servers
- Container deployment

DTTP DELIVERY PATH

GIT

GITHUB

GITHUB ACTIONS

DOCKER

AWS

10 Level 10 — Professional Software Engineering

The final level moves beyond technology.

AGILE & SCRUM

- Agile principles
- Scrum roles, events, artifacts
- Sprint planning and backlog
- Review and retrospective

PROFESSIONAL SKILLS

- Communication and requirement analysis
- Documentation and code review
- Estimation and ownership
- Teamwork, presentation, mentoring

11 Official progression

Level	Focus	Outcome
1	Web Fundamentals	Can explain how the web works
2	Frontend Engineering	Can build accessible pages and DOM interactions
3	React + TypeScript	Can build a structured SPA
4	HTTP + REST + Web APIs	Can reason about frontend-backend traffic
5	Java or Python backend	Can implement a secure REST API
6	Databases	Can design and query SQL independently
7	Architecture	Can choose structure for a reason
8	Testing, security, performance	Can protect, verify, and measure software
9	Git, Docker, CI/CD, cloud	Can deliver a running system
10	Agile + professional engineering	Can work as a team engineer

The reference material is the knowledge map. Deshmukh Technologies projects are the proof. A trainee completes this program when they can independently build, test, debug, explain, document, and maintain a full-stack application.

DTTP — DESHMUKH TECHNOLOGIES TRAINEE PROGRAM

Self-Directed Full-Stack Developer

Ten levels. One practice stack. Real projects.

Learn. Build. Solve. Review. Deploy. Grow.